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GUEST EDITORIAL COMMENT: TWO MANUSCRIPTS WITH DIFFERENT RESULTS

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I have recently become aware of a publication in the journal Tectonophysics (Seguin and Fyffe, 1986) which appears to duplicate in part a paper submitted more or less simultaneously to Geophysical Research Letters (Seguin et al., 1985). The GRL paper was accepted for publication by me during my tenure as Editor of this AGU journal. It is my purpose here to raise a few questions about these papers, in the hope that the authors will clarify the issue. I wish to note some differences in the paleomagnetic results described in these papers, which by all appearances seem to be based on the same rock units from New Brunswick, Canada. The two papers will herein be referred to as the GRL paper (Seguin et al., 1985) and the T paper (Seguin and Fyffe, 1986).

In the two publications mentioned above, the descriptions and research results are different, for what appear to be the same sites, in four ways: (1) their age assignments, (2) the observed paleomagnetic polarity, (3) the structural attitudes listed for the beds, and (4) the declinations and inclinations of the Carboniferous magnetizations obtained through laboratory analysis. The sites I am referring to are labeled R-1, B-1, B-1A, M-1 and M-2 in the GRL paper and in the same order C-1, C-3, C-2, C-10, and C-11 in the T publication. Both papers feature site location maps, so that the similarity in these two sets can be directly assessed by the reader. It should be noted that the T paper also describes results from several sites in older basalts, of Tournaisian age, which are not mentioned in the GRL paper; moreover, it must be recognized that two of the apparently similar sites (B-1A, M-1) yielded red bed results described in the GRL paper, in contrast to results (sites C-2, C-10) attributed to basalts in the T paper. Thus only three sites of basalts yielded results described in both papers (R-1 = C-1; B-1 = C-3; M-2 = C-11), but since these results are very different, with distinct and important implications for the tectonics of the region and for Carboniferous paleomagnetism in general, it seems important to query the authors about these differences.

The ages assigned to the rocks at these sites are described as Upper Namurian (B-C) - Lower Westphalian (A) in the GRL paper and as Middle Namurian (pre-uppermost Namurian) in the T paper. In both papers, Van de Poll (1973) is acknowledged as the source of this information. Since dual polarities are reported from these basalts, their

TABLE 1. Comparison of Paleomagnetic Data from Seguin et al. (GRL) and Seguin and Fyffe (T).

Site Number		Structural Correction		Polarity Assigned		Magnetic Direction			
GRL	T	GRL	T	GRL	T	GRL		T	
						D/I IS	D/I TC	D/I IS	D/I TC
Basaltic Sites									
R-1	C-1	50/15	359/15	N	R	158/32	156/19	164/12	163/11
B-1	C-3	40/ 5	22/23	R	N(M)	153/44	151/39	160/10	155/-5
M-2	C-11	60/10	0/ 0	R	R	149/35	149/25	174/-2	174/-2
Red Bed and Basaltic Sites									
B-1A	C-2	185/ 4	10/ 4	R	R	153/44	156/46	157/22	155/20
M-1	C-10	45/ 5	0/ 0	R	R	163/37	161/33	158/21	158/21

The structural correction is given as strike and dip, with the strike according to the left-hand rule (such that the strike = dip direction minus 90 degrees); polarity assigned, N = normal, R = reversed, M = mixed; D/I = Declination/Inclination (in degrees), IS = in situ, TC = tilt-corrected.

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age assignment is important for Carboniferous magnetostratigraphy and the best estimate of the onset of the Kiaman Reversed Superchron.

To facilitate comparison and a discussion of the discrepancies, I list in Table 1 the sites in common, their structural corrections, the observed polarities, and the magnetic directions as compiled from each of the two papers. In brief, it can be noted that one site (R-1 in GRL, C-3 in T) is listed in each of the two papers as revealing normal (N) polarity or predominantly normal (N/mixed) polarity, but that these two basalt sites are not at the same location. The structural corrections for the three basalt sites in common, as well as for the two localities with either red beds (GRL) or basalts (T) are minor, but clearly different. In both papers, the fold test is reported as negative, which is ascribed to non-horizontal original flow layering (i.e., primary dips).

Most importantly, however, the paleomagnetic directions, both in situ and tilt corrected, differ significantly and by much more than the reported cones of confidence. This results in an inclination difference of 27° which has important implications for the Namurian (to lower Westphalian) paleogeography of the region. The paleopoles of the two studies are likewise divergent: 21N, 135E (GRL) versus 36N, 136E (T), i.e., a difference of 15°.

Seguin and Fyffe (1986) provide sufficient details about the local stratigraphy to ascertain that the two papers indeed involve the same basaltic rocks; the difference in paleopoles, therefore, is in need of clarification. I also wish to note that the two papers do not refer to each other, although the T paper was submitted while the GRL paper was with the authors for revision. The two papers are by no means identical, but there is clear overlap in the material covered. In closing, I express the hope that the authors will explain and expand on the apparent discrepancies.

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